

Making School Handbooks Actually Useful: An AI Chat Tool Approach

Final Team Project Report

Course: Computer Science

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Project Team Members:

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Submission Context:

Computer Science

2026/5/22

We built an AI chat app to replace old, boring school rule PDFs so students, parents, and teachers can get answers to their questions instantly without scrolling forever.

1 INTRODUCTION & DESIGN PROBLEM

Let's be honest: traditional school handbooks are an absolute nightmare to read. Schools usually print them out as massive booklets or upload them as giant, disorganized PDFs hidden somewhere deep on the school website. When a student, a stressed-out parent, or a new teacher needs a quick answer—like checking if phones are allowed, finding out the penalty for a late homework assignment, or looking up what counts as an unexcused absence—they have to scroll blindly through dozens of pages of confusing law-style text. Because it takes way too long to find anything, most people just ignore the handbook completely until they accidentally break a rule and get in trouble.

Our team project fixes this problem by building the **AI Student and Family Handbook**. Instead of forcing you to scroll through a frozen text document, our app turns the handbook into a simple, interactive chat window. You just type in a normal, everyday question, and the AI instantly reads through the real handbook guidelines behind the scenes to give you the exact rule answer you need right away. This tool changes the game for how people at school find information, saving everyone a ton of time and headaches.

2 SYSTEMATIC NEEDFINDING STUDY

2.1 Research Questions and Approach

To find out how real people use school handbooks and see where they get stuck, we put together a research study. We watched people use the handbook, asked them to talk out loud while searching for rules, and sent out an online survey. We focused our research on three main questions:

- **RQ1:** How often do families actually open the school handbook, and what makes them look at it?
- **RQ2:** What specific parts of the layout or word choices cause the biggest confusion for users?
- **RQ3:** Is it easier for people to ask an AI chatbot for answers instead of flipping through a standard index table?

2.2 Quantitative Survey Insights

We gathered solid data from our online survey, which got 11 completed responses from students, parents, and teachers. The numbers showed a huge demand for an AI search tool over the current file formats.

When we asked the respondents to rate the idea of an AI-powered handbook app, the average score was an awesome **4.36 out of 5 stars**. You can see the exact breakdown of how people voted in Table 1:

Table 1: User Survey Satisfaction Ratings for AI Handbook Concept ($N = 11$)

Response Metric	Respondent Count	Sample Percentage
5 Stars (Highly Satisfied)	7	64%
4 Stars (Satisfied)	2	18%
3 Stars (Neutral)	1	9%
2 Stars (Dissatisfied)	1	9%
1 Star (Highly Frustrated)	0	0%

On top of that, we asked if they would recommend putting this AI app into their own schools. **73% (8 users)** said a clear “Yes,” **18% (2 users)** said “Maybe,” and only **9% (1 user)** said “No.”

2.3 Qualitative User Themes and Structural Pain Points

Our interviews and tests showed exactly what is wrong with the current school handbook setup:

- **The Veteran vs. New Teacher Disconnect:** We noticed a massive divide among teachers. Teachers who have worked at the school for a long time said they almost never look at the student handbook because they just use a secret internal teacher guide instead. But newly hired teachers were incredibly frustrated. They explained that trying to find clear instructions on how to handle bad behavior is super hard because the sections feel completely mixed up and out of order.
- **Demand for Situational Organization:** Every single person we interviewed (100%) said they want a digital-first experience. They specifically asked for the info to be grouped by real situations (like “I am sick at home” or “I forgot my homework”) instead of being laid out like a boring legal code block.

3 DESIGN EVOLUTION & INTERFACE ITERATIONS

3.1 Mid-Fidelity Alternatives (Check-In 2 & 3)

When we started building prototypes, our main goal was to test two completely different designs to see which layout was easier to use:

- **Interface Alternative A (Search-Centric Page):** This version was basically just a big, empty search box on the homepage. It relied entirely on the user typing their own custom prompts from scratch to track down rules.
- **Interface Alternative B (Guided AI Assistant):** This version acted more like a helpful assistant. It had clickable buttons for common school rule categories (like late homework penalties) and immediately showed clean, highlighted answer summaries right on the screen.

The complete historical mid-fidelity workspace designs, early mockups, and step-by-step layout files are in the GitHub ZIP file. You can download and check how they function and look like.

3.2 Final High-Fidelity Functional System

For our final app layout, we didn’t waste time changing the things users already liked. Instead, we focused on fixing the code to make sure the app was fully alive and connected.

First, we built the actual link between our front-end app screen and the back-end AI system. The application isn’t just a clickable mockup anymore; it actually works in real-time. Now, when you type a rule question into the chat, our AI engine reads the real uploaded handbook file to grab highly accurate, detailed answers instantly.

Second, we polished up the look of the app. We got rid of confusing instruction text blocks, adjusted the color contrast, made the buttons super obvious, and made sure the chat conversation history looks really clean and feels natural to scroll through.

The complete functional high-fidelity interactive system code, asset packs, and deployment scripts are in the GitHub ZIP file. You can download and check how they function and look like.

4 USER EVALUATION METRICS & ROADMAP

4.1 Two-Track Testing Plan

To prove our app actually does its job, we designed a balanced user testing plan divided into two tracks:

- **Track A (Qualitative Think-Aloud Interviews):** We are going to test 5 outside users (parents and students) and watch them do specific tasks. For example, we will have them try to find out the exact grade deduction for an essay turned in 48 hours late. This will let us see exactly where people get confused while chatting with the AI.
- **Track B (Quantitative Scaled Surveys):** We will share our app link with a larger group of about 20 outside participants. After they mess around with the tool, they will fill out a quick rating survey checking the app's layout, speed, and accuracy.

4.2 Critical Reflection on Internal Testing Bias

We want to be completely honest about a big limitation in our project right now: **Internal Testing Bias**. Because we had super tight deadlines and a lot of coding struggles, we have only run our tests on our own group members so far.

This is a huge problem. Since we built the app, we already know exactly how the code works, what keywords the AI likes best, and exactly where the rules are hidden inside the manual. A real student or parent opening our app for the first time will use totally different words, ask random questions, and run into roadblocks we never even thought about.

That is why our future roadmap depends entirely on getting data from external users who have never seen our project before. This will help us break out of our group bubble and get real, honest feedback.

5 REFERENCES & PROJECT APPENDIX

1. Groupmates Joe, Basco, Victor. (2026). *Check-In 1 & 2 & 3 & 4: Interface Alternative Testing, Contextual Co-Design Sessions, and Conversational Wireframes*.
2. Online System Data Questionnaire Instrument: <https://forms.gle/BxekCwmhmYqR8cPb9>.

Final Report Document